C. Earning on Bets

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1200 | Tags: binary search, combinatorics, constructive algorithms, number theory |
https://codeforces.com/problemset/problem/1979/C

You have been offered to play a game. In this game, there are n possible outcomes, and for each of them, you must bet a certain **integer** amount of coins. In the event that the i-th outcome turns out to be winning, you will receive back the amount of coins equal to your bet on that outcome, multiplied by k_i. Note that **exactly one** of the n outcomes will be winning.

Your task is to determine how to distribute the coins in such a way that you will come out ahead in the event of **any** winning outcome. More formally, the total amount of coins you bet on all outcomes must be **strictly less** than the number of coins received back for each possible winning outcome.

Input

Each test consists of multiple test cases. The first line contains a single integer t ($1 \le t \le 10^4$) — the number of test cases. The description of the test cases follows.

The first line of each test case contains a single integer n ($1 \le n \le 50$) — the number of outcomes.

The second line of each test case contains n integers $k_1,k_2,\ldots,k_n$ ($2 \le k_i \le 20$) — the multiplier for the amount of coins if the i-th outcome turns out to be winning.

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output -1 if there is no way to distribute the coins as required. Otherwise, output n integers $x_1, x_2,\ldots, x_n$ ($1 \le x_i \le 10^9$) — your bets on the outcomes.

It can be shown that if a solution exists, there is always a solution that satisfies these constraints.

If there are multiple suitable solutions, output any of them.

Example

Input
6 3 3 2 7 2 3 3 5 5 5 5 5 5 6 7 9 3 17 9 13 3 6 3 2 5 9 4 6 8 3

Output

```
27 41 12
1 1
-1
1989 1547 4641 819 1547 1071
-1
8 18 12 9 24
```

Note

In the first test case, the coins can be distributed as follows: 27 coins on the first outcome, 41 coins on the second outcome, 12 coins on the third outcome. Then the total amount of coins bet on all outcomes is 27 + 41 + 12 = 80 coins. If the first outcome turns out to be winning, you will receive back $3 \cdot 27 = 81$ coins, if the second outcome turns out to be winning, you will receive back $2 \cdot 41 = 82$ coins, if the third outcome turns out to be winning, you will receive back $7 \cdot 12 = 84$ coins. All these values are strictly greater than 80.

In the second test case, one way is to bet one coin on each of the outcomes.